Trajectory Dynamics in Self-Supervised Learning Latent Space for Audio Deepfake Detection

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***Abstract*—Human speech production is constrained by physiology, giving rise to characteristic temporal structure on acoustic signals. We hypothesise that these constraints manifest as structured trajectory dynamics in the latent space of Self-Supervised Learning (SSL) models, and that synthetic speech violates them detectably. To test this hypothesis, we train a causal Long Short-Term Memory (LSTM) next-frame predictor on bonafide speech only (Stage 1), using the deepfake-specialised SSL backbone Wav2Vec2-Large-AntiDeepfake, and compare against a static global-average-pooling baseline using identical features, thus isolating the contribution of temporal modelling. A supervised Stage 2, which trains a Multi-Layer Perceptron on the frozen LSTM internal states using labelled data, is included to characterise the role of spoof supervision. Our system achieves competitive or state-of-the-art performance across six benchmarks: ASVspoof 2019/2021, Codecfake, In-the-Wild, MLAAD-EN, and Deepfake-Eval-2024, including best published EER on ASVspoof 2021 (0.75%) and, notably, Stage 1 trained on bonafide speech only surpasses the published supervised baseline from the same backbone on DE2024 (30.35%). On near-domain benchmarks, static and dynamic approaches perform comparably. On harder cross-corpus benchmarks with diverse synthesis methods, trajectory dynamics provide substantial gains, confirming that temporal physiological constraints carry detection signal beyond utterance-level statistics.**

***Index Terms*—audio deepfake detection, self-supervised learning, trajectory dynamics, one-class learning, physiological constraints**

I. Introduction

The rapid proliferation of systems capable of producing realistic synthetic voices, whether Text-to-Speech (TTS) or Voice Conversion (VC), carries a sharp societal risk. A community has emerged to provide reliable mechanisms to detect such deepfakes, producing common benchmarks such as the pioneering ASVspoof series [1], [2] and increasingly challenging real-world evaluations [3], [4].

State-of-the-art approaches such as AntiDeepfake [5], SLIM [6], QAMO [7], and BreathNet [8] rely on SSL feature extractors such as wav2vec 2.0 [9] or XLS-R [10] as frontends. These are strong on matched benchmarks, but generalisation to diverse real-world synthesis remains an open problem: performance degrades severely on hard cross-corpus benchmarks such as MLAAD [11] and Deepfake-Eval-2024 (DE2024) [4], where the backbone representations alone are insufficient to separate bonafide from synthetic speech.

Most SSL-based systems aggregate frame-level embeddings via global average pooling or attention-weighted summarisation, discarding temporal order entirely. A smaller body of work exploits temporal structure directly: FGFM [12] selects locally suspicious frames via attention voting, TRACE [13] measures first-order embedding velocity at splice boundaries for partial deepfake detection, and BreathNet [8] guides training with explicit breath-frame annotations. However, these approaches target localised artifacts or require spoof supervision. None models the global physiological plausibility of the full utterance trajectory.

Prior work has shown that SSL latent spaces organise along patterns anchored in breath and articulation across languages [14], suggesting that physiological production constraints leave a detectable imprint in SSL geometry. We hypothesise that these constraints manifest as structured trajectory dynamics, and that synthetic speech, generated without physiological grounding, violates them detectably. To test this hypothesis, we train a causal LSTM next-frame predictor on AntiDeepfake [5] features using the bonafide-only subset of ASVspoof 2019 (Stage 1). A second stage (Stage 2) reads the frozen LSTM hidden states and trains a supervised MLP classifier on bonafide and spoof data from ASVspoof 2019. For each benchmark, we provide a controlled comparison against a static baseline trained on identical features, isolating the contribution of temporal modelling.

Our system outperforms SLIM, QAMO, and BreathNet on all shared benchmarks without backbone fine-tuning, with trajectory dynamics providing increasing advantage over the static baseline as benchmark difficulty grows — from near parity on ASVspoof to large gains on MLAAD-EN and DE2024. UMAP [15] visualisation of the SSL embedding space provides mechanistic support for this finding.

II. Related Work

A. SSL-based Deepfake Detection

Self-supervised speech representations have become the dominant frontend for audio deepfake detection, providing rich phonetic and speaker embeddings without task-specific supervision. Simple classifiers on frozen wav2vec 2.0, HuBERT, or XLS-R embeddings already deliver strong detection, confirming that these models encode rich forensic information [16]. Building on this, Ge et al. introduce AntiDeepfake [5], a series of post-trained SSL models developed on over 56,000 hours of genuine and 18,000 hours of synthetic speech. The resulting backbone exhibits strong zero-shot generalisation and serves as the frontend in our system.

B. One-Class and Anomaly-Based Detection

A recurring theme in deepfake detection is learning a compact representation of bonafide speech and flagging devia-

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tions as synthetic. OC-Softmax [17] applies one-class learning with an angular margin loss, and SAMO [18] extends this with speaker-attractor multi-centre objectives. More recently, QAMO [7] introduces quality-aware multi-centroid learning, splitting bonafide into high- and low-quality subsets based on MOS scores. All of these methods define anomaly scores from static utterance-level distances to learned centroids. Our approach shares the one-class philosophy of Stage 1 but differs fundamentally in the anomaly signal: rather than measuring distance to a static centroid, we model the temporal evolution of the bonafide trajectory and score utterances by their next-frame prediction error — capturing dynamics that static pooling discards. We note that both QAMO and our system rely on a backbone post-trained on synthetic speech [5]; the one-class claim applies strictly to the downstream trajectory model.

C. Temporal Modelling

The temporal dimension of speech has received growing attention as a complementary signal to spectral features. FGFM [12] proposes fine-grained frame selection via multi-head voting to identify locally suspicious frames, treating artifacts as spatially localised. BreathNet [8] explicitly detects breath frames and uses those locations to modulate XLS-R features during training, finding that the backbone implicitly encodes breath-related physiological cues even at inference without the breath mask — consistent with our hypothesis that SSL representations carry physiological structure. ProSDD [19] learns speaker-conditioned prosodic variation from real speech as an auxiliary objective. TRACE [13] exploits first-order embedding velocity for training-free detection of partial deepfakes at splice boundaries. Our work differs from all of these: we target full deepfakes where no splice boundaries exist, and model the global physiological plausibility of the entire utterance trajectory rather than identifying localised artifacts or measuring frame-to-frame velocity magnitude.

D. Cross-Corpus Generalisation

Generalisation to out-of-domain synthesis methods remains a central challenge [3]. SLIM [6] addresses this through style-linguistics mismatch detection, achieving competitive EER across multiple benchmarks. Yang et al. [20] improve upon SLIM through risk-aware style alignment and structural empirical risk minimisation. Recent work has further investigated backbone selection [16] and latent space augmentation [21] as strategies for OOD robustness. Our approach is complementary: rather than adapting the representation space, we model temporal trajectory dynamics within a fixed backbone, achieving lower EER than all of these systems on shared benchmarks without any fine-tuning.

III. Method

A. AntiDeepfake Features and Static Baseline

In all experiments, we extract speech features using Wav2Vec2-Large-AntiDeepfake [5], which provides 1024-dimensional frame-level representations at 50 frames per second. These features are post-trained to discriminate bonafide from synthetic speech, providing a strong representational foundation even before temporal modelling.

To quantify the contribution of trajectory dynamics independently of backbone quality, we construct a static baseline using the same features. For each dataset, we compute the centroid of all bonafide training utterance vectors and score each evaluation utterance by the L2 distance between its global average pooled (GAP) embedding and this centroid. This directly mirrors the global average pooling readout used in the original AntiDeepfake evaluation [5], discarding all temporal order information.

B. Stage 1: Causal LSTM Trajectory Predictor

We model trajectory dynamics using a 2-layer Long Short-Term Memory (LSTM) network with hidden dimension 512. Given an ordered sequence of T frame embeddings $\mathbf{h}_1, \ldots, \mathbf{h}_T \in \mathbb{R}^{1024}$ extracted from a single utterance, the LSTM causally predicts each next frame from its predecessors. The model is trained exclusively on the 2,580 bonafide utterances of the ASVspoof 2019 LA training set, minimising the mean squared error between predicted and true next frames. The resulting per-utterance anomaly score is:

$$s(\mathbf{x}) = \frac{1}{T-1} \sum_{t=1}^{T-1} \left\| \hat{\mathbf{h}}_t - \mathbf{h}_{t+1} \right\|^2 \quad (1)$$

where $\hat{\mathbf{h}}_t = f_\theta(\mathbf{h}_1, \ldots, \mathbf{h}_t)$ is the causal LSTM prediction at step t. Higher scores indicate trajectories that deviate from the bonafide dynamics learned during training.

C. Stage 2: Supervised MLP

As an optional second stage, we use the frozen Stage 1 LSTM as a feature extractor. The mean-pooled hidden state across all frames yields a fixed-length 512-dimensional utterance representation. We extract these representations for all 25,380 utterances (2,580 bonafide and 22,800 spoof) of the ASVspoof 2019 LA training set, and train a three-layer MLP with dimensions $[512 \to 256 \to 128 \to 1]$ and dropout $p = 0.3$ to output a spoof probability. Stage 2 introduces spoof supervision, in contrast to the purely one-class Stage 1.

IV. Experiments

A. Datasets

We employ ASVspoof 2019 LA train [1] as our training set, using only bonafide speech (2,580 utterances) for Stage 1, while Stage 2 additionally uses the 22,800 spoof utterances. We evaluate on six benchmarks summarised in Table I. ASVspoof 2019 LA eval and ASVspoof 2021 DF eval [2] provide controlled evaluation on studio-quality TTS and VC attacks. In-the-Wild [3] contains spontaneous celebrity speech with unknown synthesis methods. Codecfake [22] targets LLM-based neural codec synthesis. MLAAD-EN [11] is the English subset (en_US + en_UK) of MLAAD v9 with M-AILABS as bonafide source. Deepfake-Eval-2024 (DE2024) [4] is a challenging real-world benchmark of social media deepfakes; DE2024 audio is split into non-overlapping 10-second segments following the

TABLE I
EVALUATION DATASETS. †TRAINING SET FOR STAGE 1 (BONAFIDE ONLY) AND STAGE 2 (BONAFIDE + SPOOF). ‡MLAAD-EN V9 (EN_US + EN_UK); BONAFIDE FROM M-AILABS. SLIM USES MLAAD-EN V3 (37,998 UTTERANCES). §ASVSPOOF 2019/2021, CODECFAKE, AND MLAAD (SPOOF ONLY) ARE IN THE BACKBONE TRAINING DATA; ONLY ITW AND DE2024 ARE FULLY OUT-OF-DISTRIBUTION.

Dataset	Bonafide	Spoof	Hours
ASVspoof 2019 train†	2,580	22,800	$\sim$8
ASVspoof 2019 eval	7,355	63,882	$\sim$28
ASVspoof 2021 eval	22,617	589,212	$\sim$670
Codecfake§	20,000	140,000	$\sim$160
In-the-Wild	11,816	19,963	$\sim$38
MLAAD-EN‡	69,855	116,000	$\sim$220
DE2024	4,712	2,543	$\sim$21

NII evaluation protocol [5]. Notably, ASVspoof 2019/2021, Codecfake, and MLAAD spoof audio are all included in the backbone post-training data [5]; only In-the-Wild and DE2024 are fully out-of-distribution for the backbone. For MLAAD, the backbone saw only the synthetic audio; the bonafide M-AILABS speech used as our reference was not in its training set, making the bonafide trajectory model genuinely zero-shot. Prior to feature extraction, utterances are preprocessed to remove long silences exceeding 500 ms, replacing them with 200 ms natural pauses, to eliminate recording padding artifacts present in the ASVspoof 2019 corpus [1] that can act as spurious detection shortcuts [23].

B. Implementation Details

We train the Stage 1 LSTM for 500 epochs using AdamW with cosine annealing and warm restarts ($T_0 = 50$, $T_{\text{mult}} = 2$), initial learning rate 10^{-3}, dropout $p = 0.1$, and maximum sequence length 500 frames. Stage 2 MLP is trained for 100 epochs with dropout $p = 0.3$. All experiments use a single H100 GPU. Performance is reported using Equal Error Rate (EER %), consistent with the ASVspoof challenge protocol. Code and configurations are available on Zenodo [1].

To estimate variance, we train five independent Stage 1 models using 5 random seeds, and for each Stage 1 model we train five independent Stage 2 MLP classifiers, yielding 25 runs per benchmark. The static baseline is deterministic and requires no repeated runs. Reported mean and standard deviation are computed across all 25 runs, capturing both LSTM initialisation variance and MLP training variance. Stage 1 standard deviation is computed from the same 25 runs by averaging Stage 2 scores within each Stage 1 seed before computing statistics across seeds.

V. RESULTS

Table II compares our system against published baselines across six benchmarks. On near-domain benchmarks (ASVspoof 2019/2021, Codecfake), Stage 2 performs comparably to the static baseline and significantly better than Stage 1; Stage 2 outperforms all published competitors on these benchmarks. On out-of-domain benchmarks (MLAAD-EN, DE2024), Stage 1 outperforms the static baseline by a large margin and outperforms Stage 2. On DE2024, Stage 1 surpasses the NII-GAP baseline despite using no spoof supervision, while Stage 2 converges to performance similar to other supervised systems trained on the same data.

A. The Difficulty Gradient

On ASVspoof 2019/2021 and Codecfake, the gap between the static baseline and Stage 2 is small ($<$0.5% EER), indicating that the backbone representations largely suffice to separate bonafide from synthetic speech. The gap opens on In-the-Wild (-0.75% EER) where Stage 2 retains an advantage over Stage 1. On the hardest out-of-domain benchmarks, MLAAD-EN and DE2024, Stage 1 provides the best result. The advantage of Stage 1 over the static baseline grows from 17.2% on MLAAD-EN to 22.2% on DE2024, confirming that trajectory dynamics become increasingly important as benchmark difficulty grows.

This progression is directly visible in the UMAP projections of Fig. 1, computed from $\sim$175,000 randomly sampled frames per dataset using UMAP [15] with parameters $n_{\text{neighbors}}$=15, min_dist=0.1, cosine metric. ASVspoof 2019/2021 and Codecfake show a tight, well-separated bonafide cluster, consistent with their low static EER. In-the-Wild shows a larger, more diffuse bonafide distribution with partial overlap. MLAAD-EN exhibits heavy overlap, and DE2024 shows near-complete mixing of bonafide and spoof representations — explaining why the static baseline fails at 22.9% and 52.5% EER respectively. It is precisely in these harder regimes where trajectory dynamics provide the critical discriminative signal. Note also that Stage 2 variance is larger on harder benchmarks (DE2024: $\pm$2.62%, MLAAD-EN: $\pm$1.22%), reflecting sensitivity to the Stage 1 bonafide manifold estimate when bonafide/spoof overlap is substantial — consistent with the difficulty gradient observed in Fig. 1.

B. One-Class vs Supervised Generalisation

The reversal between Stage 1 and Stage 2 on harder benchmarks is informative. Stage 2 supervision is derived from ASVspoof 2019 attack types (A01–A06), which are not representative of the 140+ TTS systems in MLAAD-EN or the real-world deepfakes in DE2024. The one-class Stage 1, trained only on bonafide speech, is not affected by this mismatch and generalises more robustly to unseen synthesis conditions. Notably, on both MLAAD-EN and DE2024, Stage 2 converges to performance comparable to published supervised baselines (SLIM: 10.7% and NII-GAP: 33.36%), suggesting that the ASVspoof 2019 supervision signal, rather than the architecture, is the bottleneck for cross-domain generalisation.

VI. CONCLUSION

Our results demonstrate that trajectory dynamics in AntiDeepfake SSL space provide a reliable one-class detection signal that static pooling cannot capture, with the advantage scaling with benchmark difficulty. Moreover, the reversal from Stage 1 $>$ Stage 2 on hard out-of-domain benchmarks indicates

[1]https://doi.org/10.5281/zenodo.21879214

TABLE II

EER (%) COMPARISON ACROSS BENCHMARKS. LOWER IS BETTER. *Static* USES THE SAME ANTIDEEPFAKE BACKBONE WITH GLOBAL AVERAGE POOLING AND IS DETERMINISTIC. STAGE 1 AND STAGE 2 REPORT MEAN $\pm$ STD OVER 25 INDEPENDENT RUNS (5 STAGE 1 SEEDS $\times$ 5 STAGE 2 MLP RUNS). †NII-GAP: SAME BACKBONE WITH GAP+FC AND FULL SPOOF SUPERVISION [5]. ‡MLAAD-EN V9 (185K UTTERANCES); SLIM USES V3 (38K). §ASVSPOOF 2019/2021, CODECFAKE, AND MLAAD (SPOOF ONLY) ARE IN THE BACKBONE TRAINING DATA; ONLY ITW AND DE2024 ARE FULLY OUT-OF-DISTRIBUTION. *NII-GAP DE2024 USES WHOLE-FILE INFERENCE.

Benchmark	**Static**	**Stage 1** *(ours)*	**Stage 2**	**BreathNet** [8]	**QAMO** [7]	**SLIM** [6]	**NII-GAP**† [5]
ASVspoof 2019	1.51	2.57 ± 0.12	$\mathbf{1.11} \pm 0.10$	0.23	—	0.30	—
ASVspoof 2021	0.98	1.88 ± 0.05	$\mathbf{0.75} \pm 0.10$	1.87	1.54	3.60	—
Codecfake§	3.21	5.21 ± 0.09	$\mathbf{2.43} \pm 0.18$	—	—	—	—
In-the-Wild	4.03	4.84 ± 0.04	$\mathbf{3.28} \pm 0.60$	4.70	5.09	12.5	1.91
MLAAD-EN‡	22.86	$\mathbf{5.71} \pm 0.03$	10.02 ± 1.22	—	—	10.7	—
DE2024	52.52	$\mathbf{30.35} \pm 0.15$	35.41 ± 2.62	—	—	—	33.36*

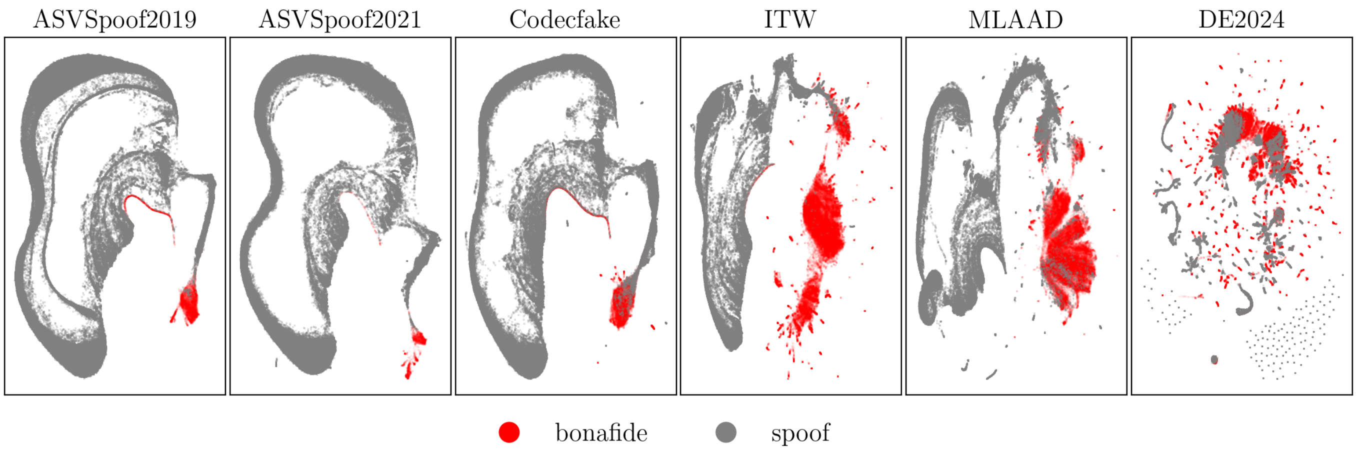

Fig. 1. UMAP visualisation of AntiDeepfake embedding space (red: bonafide, grey: spoof) across six benchmarks in increasing order of difficulty (left to right). Bonafide/spoof overlap grows progressively from tight separation (ASVspoof 2019/2021, Codecfake) through partial overlap (In-the-Wild, MLAAD-EN) to near-complete mixing (DE2024), explaining the corresponding degradation of the static baseline and the growing advantage of trajectory dynamics.

that one-class bonafide modelling generalises better than supervised spoof classification to unseen synthesis methods. We leave for future work the comparison of other sequence-based algorithms such as bidirectional LSTM or Transformers, fine-tuning with DE2024 in-domain training, and multilingual generalisation.

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